Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/25/2026
PI Name:	Okawa, Satoshi Ph.D.		
Project Title:	Investigating Space Flight-Induced Brain-Heart Axis Aging and Identifying Repurposed Drugs to Mitigate its Effects		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beheshti, Afshin Ph.D. (University of Pittsburgh)		
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	The biological effects of space flight mirror core features of accelerated aging, including mitochondrial dysfunction, chronic inflammation, and cellular senescence. These disruptions often emerge early, affect multiple organs, and can persist long after space exposure. The brain and heart – two of the most metabolically demanding and interdependent organs – are particularly vulnerable due to their tight coordination through immune, autonomic, and endocrine pathways. Yet most research considers them separately. This proposal addresses that critical gap. We hypothesize that spaceflight induces systemic, transcriptome-wide dysfunction across the brain-heart axis that reflects early stages of human aging and disease. These changes are coordinated by shared regulatory programs detectable in blood, and reversible by targeted therapy. To test this, we propose an integrated analysis of mouse and human transcriptomic data – spanning bulk, single-cell, spatial, and time-series modalities.		

Task Description:	Specific Aim 1 will use matched brain and heart transcriptomes from mice exposed to simulated microgravity, radiation, and antioxidant treatments to characterize mitochondrial and inflammatory pathway disruption. Gene regulatory networks (GRNs) will be inferred to distinguish systemic regulators from tissue-specific ones, and time-series data will help identify early-stage transcriptional drivers of brain-heart dysfunction. Specific Aim 2 will define transcriptomic senescence across both organs by mapping senescence-associated secretory phenotype (SASP) genes and a recently identified topological feature: geographical transcriptional silencing (GTS). Single-cell and spatial data will reveal cell-type and tissue-level architecture changes. Reversibility will be tested using antioxidant-treated datasets to identify drug-responsive features of senescence. Specific Aim 3 assesses whether brain-heart signatures are detectable in astronaut blood and Peripheral Blood Mononuclear Cells (PBMCs). We will test cross-tissue and cross-species conservation of space flight-induced signatures, and project them onto aging and disease atlases (e.g., GTEx – Genotype-Tissue Expression database; AMP-AD – Accelerating Medicines Partnership for Alzheimer's Disease knowledge portal) to evaluate similarity to Alzheimer's disease, cardiovascular disease, or frailty. Time-series blood/PBMC data will support analysis of molecular change dynamics post-flight. Specific Aim 4 applies a deep learning-based drug repurposing platform to identify compounds that reverse shared/systemic disruptions. Trained on tens of thousands of drug signatures, the model prioritizes interventions that restore upstream regulatory stability. In addition to benchmarking against known antioxidant treatments, we will identify drugs that act through complementary or distinct mechanisms, expanding beyond current antioxidant-based strategies. We will also evaluate whether blood-based profiles alone can inform therapeutic predictions. This study introduces key innovations: modeling aging as a coordinated multi-organ process; uncovering shared transcriptional regulators and architectural disruptions; and combining high-dimensional transcriptomics with deep learning to identify systemically acting drugs. We will also establish the relevance of space flight-induced stress to human biological aging by defining its molecular similarity to neurodegenerative and cardiovascular disease trajectories. Final outputs will include blood-based biomarkers, early regulatory targets, and candidate therapeutics with translational value for both astronaut health and age-associated diseases on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project may uncover how space flight drives brain-heart axis dysfunction through shared/systemic mitochondrial, inflammatory, and senescence pathways. Blood-based signatures of coordinated brain-heart disruption could enable non-invasive assessment of internal organ aging. By aligning these signatures with human diseases, the findings may inform early diagnostics and therapeutic strategies targeting systemic regulators, advancing precision interventions for age-related neurocardiovascular diseases, and linking physiological stress to broader mechanisms of aging.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)